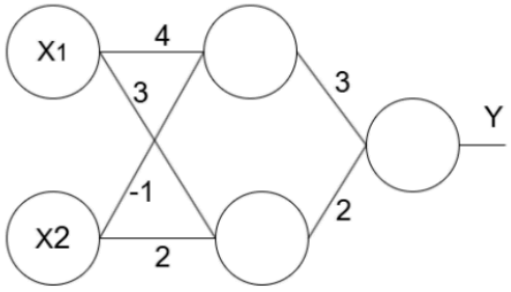
BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 422: Artificial Intelligence
MLP Problem Set

Question-01



a) If X1=5 and X2=1, apply forward propagation to calculate the value of Y. Consider that both the hidden layer and output layer is using sigmoid as the activation function
b) Mention the importance of activation functions in neural networks
c) Write the mathematical expression of the gradients for the backward propagation of the neurons in the hidden layer (As demonstrated in class)

Question-02

Consider a single neuron (perceptron) with one input x and one output y
Initial weight: w=0.4 Initial bias: b=0.1 Learning rate: η=0.1
Perform one single iteration showing forward pass, weight update and back propagation

Question-03

In this part we train a feedforward neural network with one feature.

Network Structure:

- **1 input feature,** Input: $x_1 = 1$
- **Hidden Layer 1**: 2 neurons
- **Hidden Layer 2**: 1 neurons
- **Output Layer**: 1 neuron
- Activation: sigmoid at all layers $\sigma(z) = 1/1 + e^{-z}$

A. Based on the scenario above, draw a schematic diagram of the feedforward neural network. Clearly label: each layer and its neurons, the outputs (activations) of the neurons, the flow of data through the network. [1]
B. The activation outputs from Hidden Layer 1 are 0.4 and 0.7. The weights for Hidden Layer 2 are 0.1 and 0.2. What is the activation output from the second layer? [3]